AI Search Framework v2: Executive Presentation Brief

Headline Finding: An all-SLM pipeline (≤8B) delivers **41.6% cost savings over 70B LLMs** and **88.9% savings over Frontier APIs**, cuts graph decomposition errors by **58.3%**, and closes the aggregation coherence gap by **+0.93 points** via a stylistic harmonization pass.

1. Parameter Scale Cost Benchmark (N=80)

Baseline System	Parameter Scale	Cost Ratio (SLM / Baseline)	Cost Savings (%)	95% Confidence Interval
Llama-3.1-8B	8.0B	2.656×	-165.6% (Overhead)	[2.508, 2.805]
Qwen-2.5-32B	32.5B	1.178×	-17.8%	[1.112, 1.244]
Llama-3.1-70B	**70.6B**	**0.584×**	**+41.6% Savings**	**[0.551, 0.616]**
Qwen-2.5-72B	**72.7B**	**0.584×**	**+41.6% Savings**	**[0.551, 0.616]**
Gemini-1.5-Pro	**Frontier**	**0.111×**	**+88.9% Savings**	**[0.104, 0.117]**

2. Aggregator Harmonization Benchmark (v2.2b)

- 🗣️ **Coherence Recovery:** Improved from **3.650 → 4.580 (+0.930 gain)** on 3+-domain queries.
- 🎯 **Accuracy Stability:** Correctness maintained at **4.550** and Completeness at **4.650**.
- 🔒 **Content Retention: 1.411×** retention ratio (zero content dropped).

3. Feedback Loop Structural Accuracy Gain

- **Graph Edit Distance:** Reduced from **3.57 → 1.50 (58.33% error reduction)**.
- **Economic Crossover:** Located at **≈ 35B parameters**.